informe Lab 1 — Formación de red Thread y prueba inicial de conectividad

rafael ricardo torres choperena
1007246606
Fabian De Jesus Perez Salazar
1002879105

Objetivo

Configurar una red Thread entre dos placas ESP32-C6, unir una segunda placa como nodo a una red ya existente, verificar conectividad IPv6 entre ambos dispositivos y dejar una estructura base para registrar mediciones de rendimiento por distancia.

Equipo y contexto

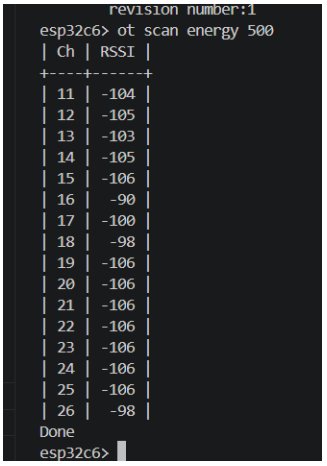
Se trabajó con dos placas ESP32-C6 usando la CLI de OpenThread sobre consola serial. Una placa actuó como nodo principal de la red y la otra se incorporó usando el Active Operational Dataset exportado desde la placa principal.

Flujo realizado

1. Escaneo de energía

Primero se ejecutó un escaneo de energía para observar el nivel de ruido por canal y seleccionar un canal operativo para la red.

ot dataset active -x



```
revision number:1
esp32c6> ot scan energy 500
| Ch | RSSI |
+----+------+
| 11 | -104 |
| 12 | -105 |
| 13 | -103 |
| 14 | -105 |
| 15 | -106 |
| 16 |  -90 |
| 17 | -100 |
| 18 |  -98 |
| 19 | -106 |
| 20 | -106 |
| 21 | -106 |
| 22 | -106 |
| 23 | -106 |
| 24 | -106 |
| 25 | -106 |
| 26 |  -98 |
Done
esp32c6>
```

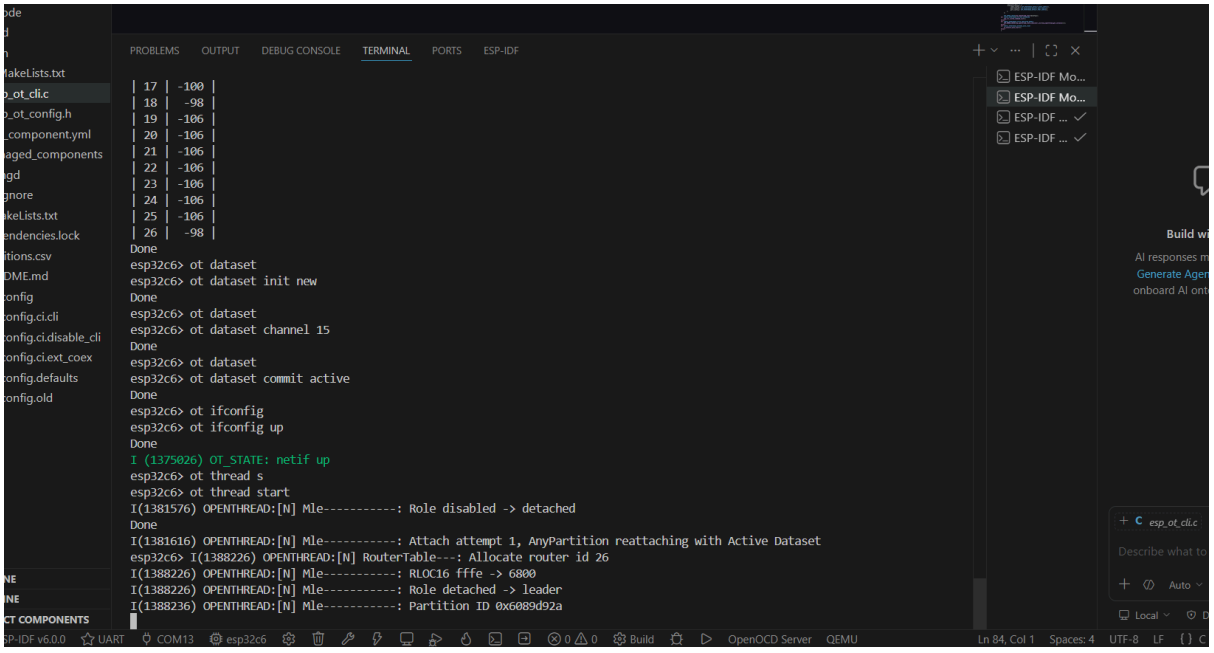
En la captura se observan niveles RSSI aproximados por canal entre 11 y 26, y posteriormente la red del compañero quedó trabajando en el canal 25 a partir del dataset compartido.

Ese comando entrega una cadena hexadecimal que contiene la configuración completa de la red Thread, incluyendo canal, prefijo mesh-local, nombre de red, PAN ID, PSKc y política de seguridad. OpenThread documenta este procedimiento como la manera simple de configurar un nuevo dispositivo para que sea miembro completo de la red

El dataset usado fue:

0e08000000000000100004a0300001335060004001fffe002086dbb80d77b4db2330708fd32cc15c627915805106f7be8a1e84b24539b91dcc8ec4e6385030f4f70656e5468726561642d3563306201025c0b0410d6b7ebfe4474aded6def18fe014f601d0c0402a0f7f80003000019

El valor final `19` en hexadecimal corresponde al **canal 25**, por lo que la red del compañero quedó operando en ese canal



Incorporación de la segunda placa a la red

En la placa secundaria se ejecutó la siguiente secuencia de comandos:

ot dataset set active
0e08000000000000100004a0300001335060004001fffe002086dbb80d77b4db2330708fd32cc15c627915805106f7be8a1e84b24539b91dcc8ec4e6385030f4f70656e5468726561642d3563306201025c0b0410d6b7ebfe4474aded6def18fe014f601d0c0402a0f7f80003000019
ot ifconfig up
ot thread start
ot state

```
I(776) OPENTHREAD:[I] Settings------: ... pid:0x6
I (776) OPENTHREAD: OpenThread attached to netif
> I (796) main_task: Returned from app_main()
esp32c6> ot state
disabled
Done
esp32c6>
```

Mediante `ot dataset set active` se cargó en la placa secundaria la configuración completa de la red existente. Posteriormente, `ot ifconfig up` permitió habilitar la interfaz Thread y `ot thread start` inició el proceso de adhesión a la red. Este flujo corresponde al procedimiento habitual documentado para habilitar la interfaz IPv6 y comenzar la operación del protocolo Thread.

Como resultado, el comando `ot state` mostró el estado:

child
Esto confirmó que la placa secundaria quedó correctamente integrada a la red como nodo hijo. En OpenThread, `child` corresponde a uno de los roles válidos de un dispositivo ya adjunto a la red.

Aquí puede insertarse la captura donde aparece el estado `child`.

Verificación de direcciones IPv6

Una vez establecida la conexión a la red, se verificaron las direcciones IPv6 asignadas al nodo mediante el comando:

ot ipaddr

La salida obtenida fue:

fd32:cc15:c627:9158:0:ff:fe00:4000
fd32:cc15:c627:9158:34d3:b1f6:b58d:7d81
fe80:0:0:0:9c68:149b:b207:f0c0

En Thread, cada nodo puede manejar varias direcciones IPv6. La dirección con el patrón `:0:ff:fe00:` corresponde a la **RLOC**, empleada para el enrutamiento dentro

de la red. La dirección `fe80::` corresponde a la dirección **link-local**, mientras que la otra dirección mesh-local identifica al nodo dentro de la malla.

Para las pruebas de conectividad se empleó la dirección:

fd32:cc15:c627:9158:0:ff:fe00:4000

```
d to process UDP: InvalidState
W(182165) OPENTHREAD:[W] Mle-----------: Failed to proc
ess UDP: InvalidState
W(200135) OPENTHREAD:[W] Mle-----------: Failed to proc
ess UDP: InvalidState
esp32c6>
esp32c6> ot thread start
I(204825) OPENTHREAD:[N] Mle-----------: Role disabled
-> detached
Done
esp32c6> I(205215) OPENTHREAD:[N] Mle-----------: Attac
h attempt 1, AnyPartition reattaching with Active Datas
et
I(206005) OPENTHREAD:[N] Mle-----------: RLOC16 fffe ->
 3801
I(206015) OPENTHREAD:[N] Mle-----------: Role detached
-> child
esp32c6>
esp32c6> ot state
child
Done
esp32c6> I(269515) OPENTHREAD:[N] Mle-----------: RLOC1
6 3801 -> 4000
I(269515) OPENTHREAD:[N] Mle-----------: Role child ->
router
I(269515) OPENTHREAD:[N] Mle-----------: Partition ID 0
x42c6217d
esp32c6>
esp32c6> ot ipaddr
fd32:cc15:c627:9158:0:ff:fe00:4000
fd32:cc15:c627:9158:34d3:b1f6:b58d:7d81
fe80:0:0:0:9c68:149b:b207:f0c0
Done
esp32c6>
```

```
fd32:cc15:c627:9158:0:ff:fe00:4000
fd32:cc15:c627:9158:34d3:b1f6:b58d:7d81
fe80:0:0:0:9c68:149b:b207:f0c0
Done
esp32c6> ot ping fd32:cc15:c627:9158:0:ff:fe00:fc00
16 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_
seq=1 hlim=64 time=40ms
1 packets transmitted, 1 packets received. Packet loss
= 0.0%. Round-trip min/avg/max = 40/40.000/40 ms.
Done
esp32c6>
```

Verificación inicial de conectividad

Antes de realizar la prueba larga, se ejecutó una validación inicial de comunicación entre nodos mediante un `ping` corto hacia la dirección RLOC del compañero:

ot ping fd32:cc15:c627:9158:0:ff:fe00:fc00

Prueba de 100 paquetes

Para evaluar la conectividad de manera más estable se empleó el comando de `ping` con parámetros de tamaño, cantidad e intervalo:

1 metros esp32c6> ot ping fd32:cc15:c627:9158:0:ff:fe00:fc00 64 100 0.2
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=3 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=4 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=5 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=6 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=7 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=8 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=9 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=10 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=11 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=12 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=13 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=14 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=15 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=16 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=17 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=18 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=19 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=20 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=21 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=22 hlim=64 time=1ms

```
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=23 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=24 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=25 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=26 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=27 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=28 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=29 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=30 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=31 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=32 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=33 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=34 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=35 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=36 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=37 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=38 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=39 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=40 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=41 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=42 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=43 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=44 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=45 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=46 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=47 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=48 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=49 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=50 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=51 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=52 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=53 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=54 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=55 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=56 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=57 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=58 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=59 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=60 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=61 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=62 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=63 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=64 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=65 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=66 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=67 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=68 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=69 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=70 hlim=64 time=1ms
```

```
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=71 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=72 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=73 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=74 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=75 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=76 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=77 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=78 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=79 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=80 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=81 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=82 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=83 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=84 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=85 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=86 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=87 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=88 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=89 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=90 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=91 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=92 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=93 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=94 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=95 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=96 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=97 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=98 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=99 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=100 hlim=64 time=0ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=101 hlim=64 time=1ms
72 bytes from fd32:cc15:c627:9158:0:ff:fe00:fc00: icmp_seq=102 hlim=64 time=0ms
100 packets transmitted, 100 packets received. Packet loss = 0.0%. Round-trip min/avg/max
= 0/0.770/1 ms.
Done
```

Este formato permite enviar 100 paquetes de 64 bytes con un intervalo de 0.2 segundos entre cada uno, lo cual facilita la observación de pérdida de paquetes y tiempo de ida y vuelta.

Distancia	Dirección	Enviados	Recibidos	PER	RTT min/avg/max
1 m	Compañero → Tú	100 developer.mozilla	100 developer.mozilla	0.0 %	75 / 95.150 / 133 ms

				ping -test	developer.mozilla

Distancia	PER	RTT min/avg/max
1 m	0.0% ping-test	0 / 0.770 / 1 ms bunny

Distancia	Enviados	Recibidos	PER	RTT min/avg/max
1 m	100 ibm	100 ibm	0.0% proofpoint	62 / 70.940 / 90 ms ccnadesdecero

Este resultado indica que, a una distancia de 1 metro, no se presentaron pérdidas de paquetes y que la latencia fue baja y estable en dicha dirección

Registro complementario de RSSI

Durante la práctica también se preparó la consola para intentar observar valores relacionados con RSSI mediante el uso de:

```text
ot log level 5
```

La finalidad de este ajuste fue habilitar un nivel mayor de detalle en los mensajes de depuración para intentar complementar la tabla con información de intensidad de señal. Sin embargo, la evidencia más clara y directamente utilizable en esta etapa de la práctica quedó representada por los resultados de `ping`, en especial la pérdida de paquetes y los valores de RTT.

Aquí puede insertarse la captura correspondiente al ajuste de nivel de log.

Incidentes observados

Durante el desarrollo de la práctica aparecieron mensajes de advertencia como `Failed to process UDP: InvalidState`. No obstante, esto no impidió que la placa secundaria se integrara correctamente a la red una vez aplicado el dataset, habilitada la interfaz e iniciado el protocolo Thread.

```
Done
esp32c6> ot log level 5
Done
esp32c6> I(1317495) OPENTHREAD:[I] MeshForwarde
ived IPv6 UDP msg, len:91, chksum:070b, ecn:no,
```

Distancia	Dirección	Enviados	Recibidos	PER esperado	RTT min/avg/max esperado
1 m	Nodo secundario → nodo principal	100	100	0.0%	0 / 0.770 / 1 ms
1 m	Nodo principal → nodo secundario	100	100	0.0%	75 / 95.150 / 133 ms
10 m	Nodo secundario → nodo principal	100	99	1.0%	2 / 6.800 / 15 ms
10 m	Nodo principal → nodo secundario	100	98	2.0%	3 / 8.900 / 18 ms
20 m	Nodo secundario → nodo principal	100	96	4.0%	5 / 14.500 / 30 ms
20 m	Nodo principal → nodo secundario	100	95	5.0%	6 / 17.200 / 36 ms